

FREDERICK REEF, Australia

WMO: 943930

Lat: 20.936S

Lon: 154.401E

Elev: 42

StdP: 14.67

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	67.1	68.0	N/A	N/A	N/A	N/A	N/A	N/A	34.6	69.5	32.5	70.1	18.2	140	0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	2.9	84.3	N/A	83.4	N/A	82.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	9.4	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
35.1	32.5	30.4	DB	64.1	86.8	1.7	1.0	62.9	87.5	61.9	88.1	60.9	88.6	59.7	89.4
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	75.7	80.1	80.4	80.0	77.6	75.2	72.6	71.2	70.8	72.3	74.0	76.2	78.7
	DBStd	3.82	1.60	1.54	1.44	1.36	1.67	1.82	1.67	1.74	1.90	1.82	1.53	1.84
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9394	933	852	930	828	782	677	656	645	670	745	786	890
	CDD65	3919	468	432	465	378	317	227	191	180	220	280	336	425
	CDH74	24445	4700	4543	4659	2830	1267	212	43	33	197	613	1736	3612
	CDH80	2161	610	718	554	26	1	0	0	1	0	0	7	244
Wind	WSAvg	17.2	16.1	17.2	19.6	20.8	18.3	18.8	18.4	16.0	14.9	15.4	15.8	14.7
Precipitation	PrecAvg	32.0	3.9	5.5	6.0	3.2	2.2	2.3	1.6	1.5	0.7	1.1	1.3	2.8
	PrecMax	48.1	10.0	19.4	16.7	10.1	9.5	6.8	8.3	5.7	2.3	5.7	8.1	9.5
	PrecMin	15.7	0.7	0.6	0.3	0.8	0.1	0.4	0.2	0.0	0.0	0.1	0.1	0.2
	PrecStd	9.1	2.8	3.7	4.8	2.6	2.0	1.7	1.8	1.4	0.6	1.2	1.7	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.2	86.0	84.6	81.0	80.1	78.5	76.3	76.4	77.8	79.6	80.9	84.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	83.9	84.4	83.5	80.8	79.1	77.1	75.1	74.8	76.8	78.2	80.0	83.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	83.1	83.3	82.8	80.3	78.3	76.0	74.4	73.8	76.0	77.4	79.2	82.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	82.5	82.6	82.3	79.8	77.6	75.2	73.8	73.2	75.2	76.8	78.5	81.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	2.8	2.9	2.6	2.6	2.6	2.9	2.6	2.6	2.6	2.5	2.4	2.7
		MCDBR	3.7	3.7	3.3	3.1	2.8	3.7	3.1	3.4	3.6	3.5	3.4	4.0
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.383	0.385	0.380	0.357	0.341	0.346	0.323	0.330	0.342	0.364	0.371	0.373	
	taud	2.592	2.605	2.613	2.648	2.649	2.626	2.680	2.645	2.588	2.546	2.557	2.599	
	Ebn at Noon	305	300	294	289	282	274	285	293	301	303	307	308	
	Edh at Noon	34	33	31	28	26	25	25	28	31	34	35	33	
All-Sky Solar Radiation	RadAvg	2244	2077	1814	1631	1378	1187	1324	1627	1992	2219	2349	2347	
	RadStd	127	203	173	108	82	96	61	93	67	128	168	169	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page